M-Swin: Transformer-Based Multiscale Feature Fusion Change Detection Network Within Cropland for Remote Sensing Images

Jun Pan, Yuchuan Bai, Qidi Shu, Zhuoer Zhang, Jiarui Hu, and Mi Wang, *Member, IEEE*

***Abstract*— Remote sensing image change detection is extensively utilized in various applications in the field of remote sensing, particularly in the realm of cropland conservation, where it plays a critical role in protecting the agro-ecosystem and ensuring global food security. However, the progressive improvement in resolution and size of remote sensing imagery has led to a "scale gap" challenge in the detection of small building changes in cropland areas. To address this challenge, an innovative multiscale feature fusion change detection network (M-Swin) based on transformer using hierarchical windows is proposed. In order to obtain clearer edges and better separation of the change results, a novel Siamese transformer encoder (MSW encoder) is proposed, which can better capture the change information in small building through hierarchical windows and fuse the multiscale feature obtained from different windows. To effectively reduce missed and misdetected small area of changing buildings, a novel bitemporal image feature fusion module (BFFM) is proposed, which can enhance the features based on a priori guidance, thus improving the saliency of change regions. Additionally, a new remote sensing image change detection dataset for cropland, called LuojiaSET-CLCD, has been proposed, experimentally demonstrates that M-Swin has good potential for highly accurate change detection of small buildings within cropland areas and outperforms several newly existing methods in three datasets (LEVIR, WHU-CD, and LuojiaSET-CLCD). Our dataset will be publicly available at https://github.com/RSIIPAC/LuojiaSET-CLCD.**

***Index Terms*— Change detection, deep learning, remote sensing images, Siamese network, small change buildings.**

I. Introduction

CHANGE detection in remote sensing imagery aims to extract the areas of change of interest within bitemporal images. It has been widely used in various fields, such as land use [1], urbanization [2], and natural disaster investigation [3], and has achieved remarkable achievements. The increasing occupation of cropland due to human activities is having a negative impact on local agroecosystems and global food security [4]. Therefore, how to accurately extract the ground change areas within the cropland in bitemporal remote sensing images has become a hot issue in the field of remote sensing. At present, many scholars have made a series of researches in the field of change detection, and the existing change detection methods can be broadly classified into two categories: traditional change detection methods and deep learning change detection methods.

The pixel-based change detection model was the first method proposed for change detection and determines the areas of change by analyzing spectral differences between individual pixels. Methods like image difference and image ratio [5] calculate change by thresholding algebraic calculations. Change vector analysis (CVA) [6] is another algorithm used for detecting changes in bitemporal images, and various scholars have proposed enhancements to it, such as compressed CVA (C^2VA) [7], robust CVA (RCVA) [8], and logically verified change vector analysis [9]. Other methods use image projection transformation and adjustment of band combinations to increase the detection accuracy of change, such as principal component analysis (PCA) [10], [11], multivariate alteration detection (MAD) [12], [13], and slow feature analysis (SFA) [14], [15]. However, traditional pixel-based methods can only detect changes within a single pixel, leading to fragmentation and pretzel noise in detection results. To solve this problem, researchers have attempted to use spatial contextual information for object-level change detection. Object-based change detection models first delineate homogeneous regions and then detect change regions using objects as detection units [16], [17], [18], [19]. Conventional methods strongly depend on object segmentation accuracy and are insufficient for complex ground conditions in multitemporal high-resolution imagery [20].

Recently, driven by the massive amount of remote sensing data, a large number of methods using deep learning techniques for change detection have emerged. Benefits from the powerful information mining and feature characterization capabilities of deep learning, the deep learning-based change detection methods can simultaneously extract features from bitemporal images and automatically discriminate change regions. Recurrent neural network (RNN) [21], Siamese convolutional neural network [22], and recurrent convolutional neural networks [23] have been used in change detection

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methods for remote sensing images, which have produced good detection results. However, these patch-based change detection methods can only obtain change or nonchange labels at the patch level and are time-consuming when performing pixel-by-pixel inference tasks.

In contrast to patch-based methods, Daudt et al. [24] first applied a fully convolutional network (FCN) to the change detection task, which can accept image inputs of different sizes, saving a significant amount of time while completing the annotation of each pixel. Lei et al. [25] proposed a U-shaped structure of FCN for landslide detection and used a pyramid pooling module [26] to acquire multiscale information. Inspired by UNet++, Peng et al. [27] improved the structure of UNet++ to learn multilevel features in high-resolution images. To better fuse the features of bitemporal images, many attention mechanisms have also been applied. Chen and Shi [28] designed a self-attention mechanism to model the spatiotemporal relationships of bitemporal images, highlighting the regions of change. Zhang et al. [29] obtained more accurate change boundaries by computing channel attention and spatial attention for the fusion of features at different levels. Shu et al. [30] promoted the integrity of the change object by extracting change information from different views of the bitemporal image and using the change context module, while improving semisupervised change detection performance by using a single temporal architectural prior as a supervisory signal [31].

The transformer-based networks [32] (e.g., VIT, Swin, etc.), which benefits from its unique self-attention mechanism, have achieved SOTA accuracy in various computer vision tasks [33], [34]. BIT [35] tokenizes bitemporal image information and applies transformer encoder to model contextual information. The resulting global token is fed back to refine original features using transformer decoder. ChangeFormer [36] employs a hierarchically structured transformer encoder alongside a multilayer perception (MLP) decoder in a Siamese structure to present a comprehensive view of multiscale details, which is imperative for precise change detection.

However, the development of remote sensing satellite technology has led to an increase in the size and resolution of remote sensing images, resulting in a "scale gap" between individual features and the entire image. This gap has become a significant challenge for the intelligent interpretation of high-resolution remote sensing imagery, particularly for change detection tasks. Compared to other tasks, the unique semantic label of "change" makes change targets more sparse, particularly within the cropland area where small change areas are prevalent. Nowadays, the detection of small change buildings is still facing serious challenges, which can be described in detail by following two scenarios.

Scenario 1: When multiple small change buildings are present, their boundaries can become blurred, and individual targets may be difficult to distinguish from one another. This issue is exemplified in Fig. 1(a), where the ground truth of the change target is represented by white, and the detection result is represented by red. The existing methods may not accurately capture the fine boundary information, causing the boundaries of multiple targets to merge.

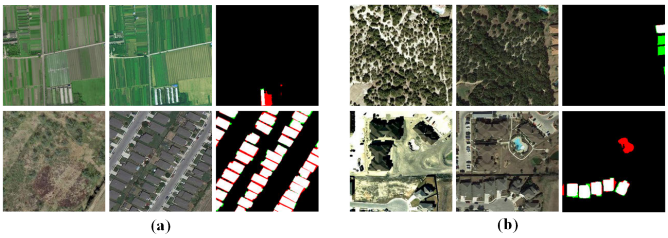


Fig. 1. Challenges of small change target detection within the cropland area. (a) Unclear edges and blurred boundaries. (b) Targets be missed or misdetected.

Scenario 2: Small change buildings within the cropland area are more likely to be missed or misdetected. As shown in Fig. 1(b), in scenarios where the area of homogeneous lands, such as croplands and plains, is relatively large, small change buildings (e.g., small buildings) among them are easily misclassified as unchanged areas (green areas) due to the category imbalance problem. At the same time, there are also features that are incorrectly classified as changed areas (red areas) due to seasonal and human factors.

In order to deal with these challenges better, this article proposes an innovative transformer-based multiscale feature fusion change detection network (M-Swin) with a hierarchical window for detecting small change buildings within the cropland area. The network fuses multiscale information perceived by the hierarchical window to obtain richer boundary detail features, resulting in clearer edges and better separation of change results. By enhancing the features of change and invariant regions separately in the bitemporal images, the network improves the saliency of change targets and effectively reduces the number of missed or misdetected small change buildings within the cropland area. Furthermore, a new remote sensing image building change detection dataset called LuojiaSET-CLCD is proposed. This dataset contains a large amount of data facing challenging scenarios and is particularly suitable for the detection of small building changes within the cropland area.

The forthcoming sections of this article are presented in the following order. Section II gives a concise overview of the prior research related to the content of this article. Section III presents the primary framework of M-Swin, along with two innovative modules and their corresponding details. Section IV encompasses the experimental analysis, wherein the efficacy of the proposed innovation modules is demonstrated through feature visualization, and the performance of M-Swin is validated on three datasets. In Section V, this article is concluded, and an outlook on future work is provided.

II. Previous Works

A. Siamese Convolutional Neural Network

The Siamese convolutional neural network is a special network formed by splicing two networks with the same network structure and shared weights, which can extract the feature information of two input images in the same distribution domain, avoiding the problem of domain differences generated

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by using two different networks and realizing the similarity metric between two images.

In the change detection task, the Siamese network architecture [24] does not use the traditional patch-based strategy, improves the speed and accuracy of the prediction phase, and is able to process the input images of any size with sufficient memory. The main difference between the Siamese network architecture and the early fusion-based method is that the early fusion-based method concatenates the two images before the network input and treats the two images as different channels, while the Siamese network architecture processes the two images separately through two network branches with the same network structure and shared weights and then fuses the two branches through subsequent convolutional layers. The Siamese convolutional neural network processes the two images separately in a way that provides the network with an explicit comparison between the two image features and is suitable for network design for change detection tasks.

B. Multiheaded Self-Attentive Mechanism and Transformer

The self-attention mechanism is the basic unit of the self-attention module in transformer [32] and the basic idea of information extraction implemented in transformer. It can be represented as follows:

$$\begin{aligned} \mathbf{Q} &= \boldsymbol{W}_q \boldsymbol{X}; \quad \mathbf{K} = \boldsymbol{W}_k \boldsymbol{X}; \mathbf{V} = \boldsymbol{W}_v \boldsymbol{X} \\ \mathbf{Attn} &= \text{softmax}_{\text{dim}=-1}\left(\frac{\boldsymbol{Q}\boldsymbol{K}^T}{\sqrt{d}}\right) \\ \boldsymbol{Y} &= \mathbf{Attn} \times \mathbf{V}. \end{aligned} \tag{1}$$

First, three dimension-preserving linear transformations are performed on the input $\boldsymbol{X}$ to obtain $\mathbf{Q}$(query), $\mathbf{K}$(key), and $\mathbf{V}$(value), respectively. Subsequently, the attention matrix **Attn** is obtained by multiplying the matrix $\mathbf{Q}$ with the matrix $\mathbf{K}$. The attention matrix is normalized by using $\sqrt{d}$ to correct the variance of the matrix elements to 1. The row vectors of the matrix are normalized by using softmax to obtain each $\mathbf{V}$ weights. Finally, the matrix multiplication of **Attn** and $\mathbf{V}$ is done to obtain the result $\boldsymbol{Y}$ after correlation calculation

$$\mathbf{O} = \mathbf{W}\left[\boldsymbol{Y}^1, \ldots, \boldsymbol{Y}^i\right]^T. \tag{2}$$

In order to capture multiple correlations between vectors, multiple vectors $\mathbf{Q}$ and $\mathbf{K}$ need to be introduced, and then, multiple computations of self-attentiveness are performed, and the results $\boldsymbol{Y}^1, \ldots, \boldsymbol{Y}^i$ (i denotes the number of heads) are stitched together, and finally, the stitched matrix $\mathbf{Y}$ is converted to its original size by a transformation matrix $\mathbf{W}$ to obtain the final output $\mathbf{O}$.

Self-attentive mechanisms are widely used in various transformer-based methods, but this self-attentive computation results in excessively large computational effort. Since the transformer was first used in computer vision tasks, some scholars have made improvements to address its computational effort. PVT [37] introduces the traditional CNN pyramid structure, uses convolutional layers between stages to reduce the number of tokens, and adds additional convolutional kernels to reduce the feature map size when computing $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$. Swin [38] reduces computation by a strategy of computing self-attention within a fixed window. Twins [39] uses relative position coding instead of absolute position coding while using a local–global self-attention mechanism alternatively to reduce the computational redundancy.

III. Method

The M-Swin proposed in this article can better capture small area change information through a hierarchical window and is guided by a priori information for feature enhancement, resulting in optimized detection results. The network framework, shown in Fig. 2, follows a U-shaped structure and uses a classical encoder–decoder architecture found in convolutional neural networks. It consists of three main components: the MSW encoder, bitemporal image feature fusion module (BFFM), and a transformer decoder with deep supervised multilevel feature fusion.

In the proposed M-Swin framework, the bitemporal RGB images I_1 and I_2 ($H \times W \times 3$) of the same region are cropped into nonoverlapping patches, which are concatenated to form a token sequence ($H/4 \times W/4 \times 48$) and fed into the MSW encoder for multiscale multilevel feature extraction. The resulting hierarchical output is similar to that of a CNN encoder. Next, the bitemporal multiscale features from each level of the Siamese MSW encoder are input to DFFM for feature enhancement and then combined into bitemporal fusion features. Finally, the multilevel bitemporal fused features are fed into each level of the decoder for multilevel feature fusion. The resulting features are gradually restored to the original image size using several Swin blocks, and the predicted change maps of different depths obtained from each deep supervisory branch are aggregated to form the final change map.

A. Feature Extraction Encoder (MSW Encoder) With a Hierarchical Window

To address the issue of blurred boundaries and unclear edges of small change buildings, a novel Siamese transformer encoder (MSW encoder) with a hierarchical window for extracting shared weights of multiscale features of bitemporal images is proposed. The MSW encoder can better capture small-area change information through hierarchical windows and obtain change results with clearer edges and better separation by fusing multiscale feature information obtained from multiple windows. The structure of the MSW encoder is illustrated in Fig. 3, and it mainly consists of multiscale windowing, W-MSA/SW-MSA, residual connections, and multiscale feature fusion. It can be represented by

$$F_{\text{output}} = \text{MLP}\left\{\text{Concat}\left(\text{Swin}_1\left(F_{\text{input}}\right), \ldots, \text{Swin}_i\left(F_{\text{input}}\right)\right)\right\} \tag{3}$$

where F_{input} and F_{output} denote the input and output of MSW block, Swin represents Swin block, i denotes the number of layers of hierarchical window, Concat denotes the concatenate operation, and MLP denotes the multilayer perceptron.

1) Patch Embedding and Patch Merging: Unlike the CNN-based approach, M-Swin requires the conversion of image information from two dimensions to one dimension by patch

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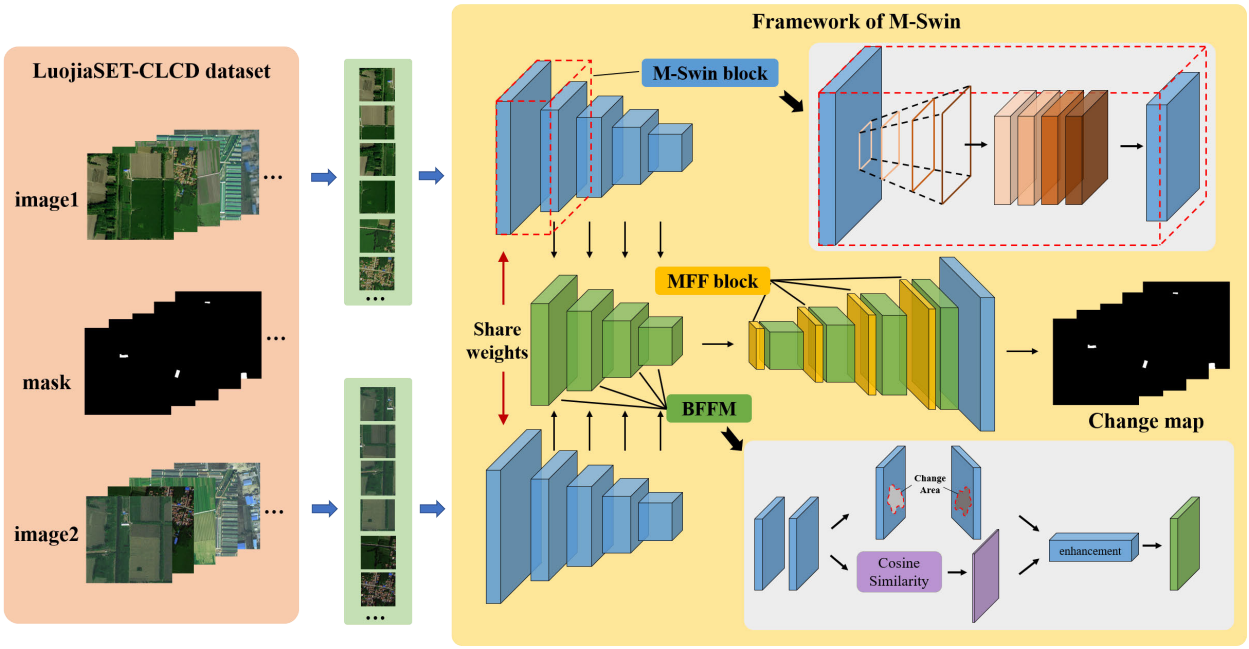


Fig. 2. Overall framework of M-Swin.

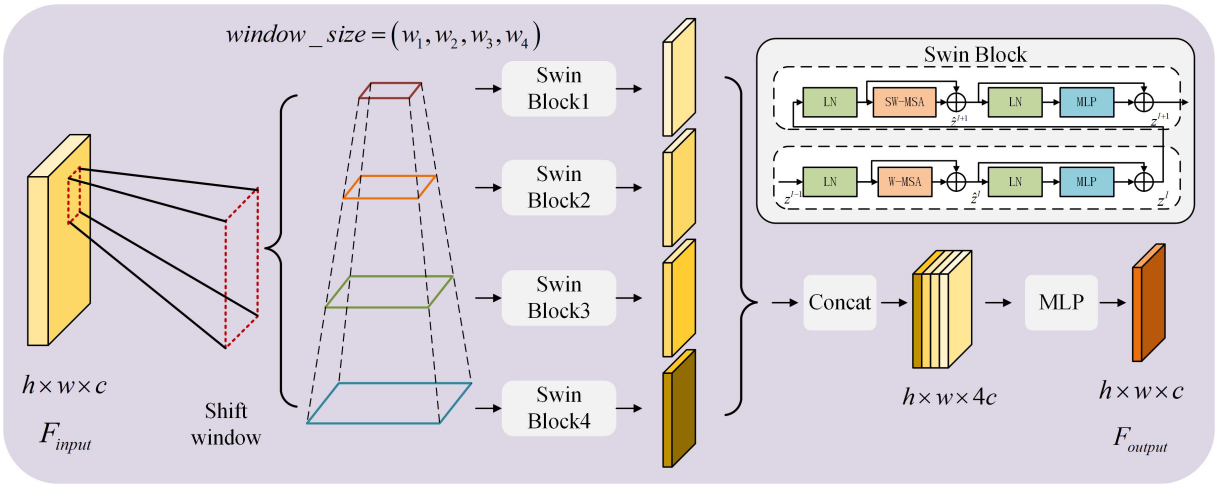


Fig. 3. Multiscale-Swin transformer block with four windows.

embedding and downsampling of the feature map using patch merging.

The patch embedding layer exists only in stage1 and is the first layer of the whole encoder. For the input token sequence, the patch embedding layer first performs a dimensionality reduction by linear embedding, merging the dimensions of H and W to make a projection from two dimensions to one dimension, which is then fed into the MSW block.

The patch merging layer exists after each MSW block and it downsamples the features extracted by the MSW block to reduce the feature resolution, thus producing hierarchical multilevel features similar to a typical CNN encoder. The size of the downsampled feature map will be 1/2 of the original size and the number of channels output will be controlled by a linear layer to be twice the original size.

2) MSW Encoder Block: Remote sensing images have unique multiscale characteristics compared to natural images. For example, the same type of target also appears at different scales on the same remote sensing image. Some large buildings can easily exceed the size of smaller buildings by a factor of several dozen.

The multiscale characteristics of remote sensing images often make it more difficult to detect changes in small buildings within cropland. For this problem, different types of convolutional kernel groups can provide a feasible solution for conventional CNNs [40], [41], which provides important inspiration for this article. Different types of window groups can control the size of the receptive field like different types of convolutional kernels. According to this, the MSW block optimizes the feature extraction algorithm by designing hierarchical windows of different sizes in different depths of the

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network, calculating multiheaded self-attention under these windows, and finally fusing the calculation results of each layer to achieve multiscale feature extraction.

For a given input, the biggest difference between MSW block and Swin block is that MSW block sets up a hierarchical window to capture the multiscale feature information of the image according to the level it is on and the size of the input. Our intuition is that larger windows are suitable for extracting globally structured features, while smaller windows are suitable for extracting locally refined features, and that the number of types of hierarchical windows is related to the size of the input feature map. It can be reasonably inferred that shallow features require more window types than deep features for extracting multiscale feature information. This is because shallow features represent the structural and textural information of an object at a higher resolution, and using more window types can extract more comprehensive information. On the other hand, deep features represent the semantic information of the object at a smaller resolution, and using as many window types as shallow features would lead to redundancy and computational burden. Hence, M-Swin divides each stage into windows of different scales and reduces the number of window types as the network goes deeper.

As shown in Fig. 3, the Swim block is a part of the MSW encoder and its computation can be represented as

$$\mathrm{Swin}\left(F_{\mathrm{input}}\right) = \mathrm{MLP}\left(\mathrm{SWMSA}\left(\mathrm{MLP}\left(\mathrm{WMSA}\left(F_{\mathrm{input}}\right)\right)\right)\right) \tag{4}$$

where F_{input} denotes input, MLP denotes multilayer perceptron, WMSA denotes calculating multiheaded self-attention in a window, and SWMSA denotes calculating multiheaded self-attention in a shifted window.

MSA is a common module in various transformer methods, and we have described its structure and calculation in detail in our previous work. While it enables the extraction of feature information through multiple global self-attention calculations, it also imposes a significant computational burden, and its computational complexity being a multiple of the square of the input size.

Since the division of multiple same-level windows does not overlap, calculating self-attention only within a window brings about a disconnection of information between multiple windows. As shown in Fig. 4, in order to increase the information interaction between adjacent windows, M-Swin uses the SW-MSA module to introduce the shifted window operation to achieve information interaction with other windows. Since M-Swin has different sized hierarchical windows in each stage, the SW-MSA module computes differently, calculating multiple heads of attention on the same target in multiple windows at the same time, thus obtaining information on the features of the same target at different scales and fusing them, SW-MSA in M-Swin is also equivalent to W-MSA with the same number of windows.

As shown in Fig. 4, for the shifted window strategy in M-Swin, in order to more intuitively feel the change of window size, this article shows an 8×8 patch in the area of the illustration. Since M-Swin adopts a window group with different window sizes, this article gives the window divisions

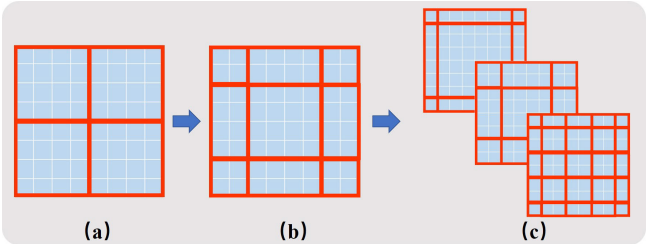


Fig. 4. Shifted window strategy in M-Swin within 8×8 patches. (a) Window division in W-MSA. (b) Window division in SW-MSA. (c) Window division in SW-MSA of M-Swin.

of three different window sizes, whose window sizes are 6, 4, and 2. Due to the difference in window sizes, the 8×8 patches are only a localized situation, and in fact, the total number of windows is determined by the size of the input image, the patch size, and the window size.

Like many CNN network modules, to address the difficulty of training multilayer neural networks, Swin blocks have adopted the residual connection structure. MLP is used to control the number of output channels in multiscale feature fusion, in addition to enhancing the expressiveness of the model after each W-MSA/SW-MSA operation. After passing through multiple Swin blocks with different scale windows, the extracted multiscale features are first concatenated together and then restored to the number of channels of a single feature by an MLP to complete the multiscale feature fusion.

B. BFFM Based on Priori-Guided Feature Enhancement

To tackle the challenge of small change buildings being missed or misdetected, a novel BFFM that utilizes priori-guided feature enhancement is proposed. As missed and misdetected change targets manifest themselves at the feature level as pseudo-invariant and pseudo-change features, BFFM can help the model to better determine them accurately. With the help of the a priori information of cosine similarity, BFFM achieves the suppression of pseudo-change features by enhancing features in regions that may have changed and the suppression of pseudo-invariant features by enhancing features in regions that may not have changed. This approach improves the saliency of change regions and significantly reduces the likelihood of small change buildings being missed or misdetected.

The structure of the BFFM is shown in Fig. 5, and it can be represented by

$$\begin{aligned}
\mathrm{Cossim} &= \mathrm{Cos}(F_1, F_2) \\
F_{\mathrm{Absub}} &= |F_1 - F_2|; \quad F_{\mathrm{Add}} = F_1 + F_2 \\
F^{\mathrm{CE}} &= F \times |\mathrm{Cossim} - 1|; \quad F^{\mathrm{IE}} = F \times (\mathrm{Cossim} + 1) \\
F_D &= \mathrm{MLP}\left\{\mathrm{CA}\left(\mathrm{Concat}\left(F_1^{\mathrm{CE}}, F_1^{\mathrm{IE}}, F_2^{\mathrm{CE}}, F_2^{\mathrm{IE}}, F_{\mathrm{Absub}}, F_{\mathrm{Add}}\right)\right)\right\}
\end{aligned} \tag{5}$$

where F_1 and F_2 denote the bitemporal features of the input, F_D denotes the fused features of the output, Cos denotes the cosine similarity calculation, F^{CE} denotes the features with change region enhancement, F^{IE} denotes the features with invariant region enhancement, Concat denotes the concatenate

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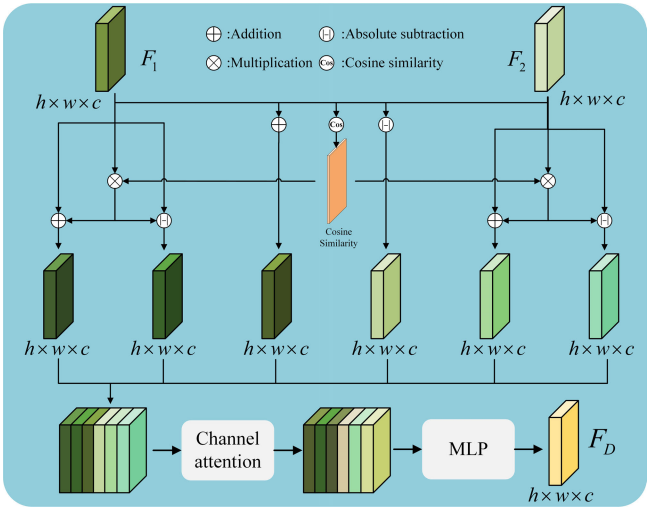


Fig. 5. BFFM based on priori-guided feature enhancement.

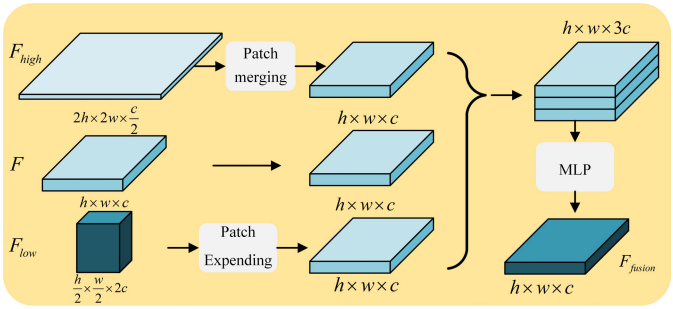


Fig. 6. MFF block.

operation, CA denotes the channel attention module, and MLP denotes the multilayer perceptron.

1) Temporal Coherence of Bitemporal Images: For the bitemporal images input to the network, our intuition is that the resulting change map should remain the same even if network swaps the before and after order of the two images. To achieve this, BFFM uses three computational methods to extract difference features from the output bitemporal image feature information: addition, absolute value subtraction, and cosine similarity calculation. The cosine similarity calculation treats the bitemporal feature map as two vectors and measures their similarity by calculating the cosine of the angle between them. This approach helps to maintain temporal consistency while ensuring that small changes are captured effectively.

2) Priori-Guided Feature Enhancement: Change features are indeed crucial in change detection, but for binary classification tasks, enhancing invariant region features is equivalent to enhancing change region features, so invariant features should be on an equal footing with change features, and equal attention should be paid to invariant and change features.

Considered the fact that cosine similarity can reveal the similarity between bitemporal feature maps, a novel priori-guided feature enhancement method is proposed that enables the model to accurately determine the pseudo-invariant and pseudo-change features and detecting targets that have actually changed. First, we compute the cosine similarity of the bitemporal feature map and use it as a priori to multiply the corresponding element of the input feature map, resulting in an activated feature map. This map preserves the original feature structure and enhances the difference between the change and invariant regions within a single feature map. The activated feature map is characterized by positive eigenvalues for constant regions and negative eigenvalues for changing regions, with the magnitude of the eigenvalues decreasing with the degree of change. To enhance the invariant region, we add the activation feature map to the original features, and to enhance the change region, we subtract the activation feature map from the original features.

It is worth mentioning that the feature enhancement for the changed or invariant regions is not sequential, but independent of each other.

3) Bitemporal Feature Fusion: To preserve as much original feature information as possible and minimize feature loss, BFFM begins by concatenating the six processed feature maps. Then, assign weights to these feature maps using channel attention, enabling the network to emphasize important features and suppress less important ones. Finally, enhance the model's expressiveness with an MLP module while controlling the output channel count.

C. Transformer Decoder Based on Deep Supervised Multilevel Feature Fusion

As shown in Fig. 6, unlike the Swin decoder, M-Swin decoder replaces the original patch expend layer with MFF to achieve the fusion of different resolution features. In addition, a deep supervision strategy is applied to the decoder to produce different layers of predicted change map output, increasing the reliability of the result.

1) Multilevel Feature Fusion: The hierarchical multilevel feature output produced by MSW encoder plays a crucial role in the reconstruction of the decoder information. In particular, the high-resolution shallow features help to predict the structure in the change map, boundary information, and information reconstruction for small change buildings, while the semantic information contained in the low-resolution deep features is the basis for distinguishing whether the location has changed or not.

The MFF module receives the feature map at the same depth while fusing feature information from the previous level with the next level. The size of the higher resolution feature map of the previous level and the lower resolution feature map of the next level are converted to the same size as the feature map of the current level by means of patch merging and patch expend; then, the feature maps of the different levels are concatenated together and fed into an MLP to reduce the number of output channels and enhance the expressiveness of the model. Finally, the fused features go through a patch expend layer to gradually expand the feature size and are fed into the next Swin block.

2) Transformer Decoder With a Deep Supervision Strategy: In general, deepening the depth of a neural network can improve the representational power of the network at the cost of making it difficult to train. Deep supervision is a common solution to alleviate the problem of hard training of deep

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neural networks and solves the problems of vanishing training gradients and slow convergence by adding some auxiliary branching classifiers to some intermediate layers of the deep neural network. For each of the four different depths of the decoder output, it was passed through the final MLP head for pixel-by-pixel change detection to obtain the final predicted change maps for the different depths.

IV. Experiments and Discussion

To effectively validate and evaluate the proposed M-Swin, a comprehensive experiment was conducted on three datasets: two publicly available datasets (LEVIR and WHU-CD) and one newly proposed dataset (LuojiaSET-CLCD). We compared the experimental results of M-Swin with five state-of-the-art methods (FC-Siam-conc, FC-Siam-diff [24], STANet [28], BIT [35], ChangeFormer [36], SwinSUNet [42], and RDP-Net [43]) and performed ablation experiments for each innovative module. The effectiveness of the innovative modules and the superiority of M-Swin were verified using various evaluation metrics, such as precision, recall, F1, and IoU. Additionally, the feature map is visualized to further elaborate the role of the innovation module, and the experimental results of each dataset are also discussed and analyzed.

A. Datasets

1) LuojiaSET-CLCD Dataset: The detection of changes in remote sensing data is a critical aspect of remote sensing applications. While deep learning-based change detection methods for remote sensing images have made significant strides in recent years, the availability of datasets for scholars remains extremely limited, greatly restricting the innovation of change detection methods and their applications in remote sensing. To address this issue, a new cropland change detection dataset (LuojiaSET-CLCD), characterized by a large quantity of data, high resolution, and challenging scenarios that are suitable for detecting changes within cropland is proposed. It is worth mentioning that LuojiaSET-CLCD has been provided to the 2022 International Remote Sensing Image intelligent Processing Algorithm Contest (2022 RSIIPAC), supported by the International Society for Photogrammetry and Remote Sensing (ISPRS), as a contest dataset. We hope that the LuojiaSET-CLCD dataset will contribute to the innovation of cropland change detection methods and their corresponding applications.

The LuojiaSET-CLCD dataset contains 4194 pairs of high-resolution bitemporal remote sensing images, measuring 512 × 512 pixels, from GF1 and GF2, with an optical resolution of 1–2 m. These images were taken in 2016–2018 and cover various typical scenes, such as urban areas, plains, arable land, and mountainous areas, in different seasons. The types of change targets mainly include the addition and demolition of buildings and construction areas. The LuojiaSET-CLCD dataset stands out from other datasets in two significant ways. First, it features change targets of various scales, with large and small targets coexisting, making it more challenging. Second, it not only focuses on changes to erected buildings but also labels buildings that have not yet been constructed. Considering that the density of buildings in urban areas is gradually increasing, the area of buildings, especially illegal buildings, is also becoming smaller, and that the construction and demolition of buildings is a dynamic and continuous process, it is obvious that the addition of the detection of small buildings and uncompleted buildings is more relevant to the dynamic monitoring of land use, the investigation and monitoring of mineral resources development and geological hazards, and the survey of the quantity and quality of arable land.

2) Other Datasets: In addition to the LuojiaSET-CLCD dataset, M-Swin is validated using two other publicly available datasets commonly used in the field of change detection (LEVIR and WHU-CD).

The LEVIR dataset consists of 637 pairs of high-resolution bi-temporal Google Earth images, each with a size of 1024 × 1024 pixels and an optical resolution of 0.5 m. The dataset focuses on building growth and deletion within the past 5–14 years. On the other hand, the WHU-CD dataset comprises two aerial images of size 15 354 × 32 507 pixels with a resolution of 0.2 m. It provides a change map for the corresponding area in raster format, which can be cropped and divided according to the researcher's requirements.

B. Evaluation Indicators

In order to validate the performance of the proposed model, we used several evaluation metrics commonly used for change detection: precision, recall, F1, and IoU. The corresponding calculations are as follows:

$$\begin{aligned} \text{Precision} &= \frac{\text{TP}}{\text{TP}+\text{FP}} \\ \text{Recall} &= \frac{\text{TP}}{\text{TP}+\text{FN}} \\ \text{F1} &= \frac{2\times\text{Precision}\times\text{Recall}}{\text{Precision}+\text{Recall}} \\ \text{IoU} &= \frac{\text{TP}}{\text{TP}+\text{FN}+\text{FP}} \end{aligned} \tag{6}$$

where TP indicates true positive, FP indicates false positive, and FN indicates false negative.

C. Implementation Details

Our model was built based on Pytorch and trained on four GeForce RTX 3090 GPUs. In our experiments, we loaded the Swin transformer trained on the Imagenet dataset as a pre-trained model and then performed migration learning on three change detection datasets. During training, we performed data augmentation on the training data, including random flipping, random scaling, random cropping, random color perturbation, and Gaussian blurring, while we used a weight-decaying AdamW optimizer and a CE loss function. The AdamW optimizer weight was set to 0.01 and the beta value was set to (0.9, 0.999), and we also used a learning decreasing rate strategy, where the learning rate of the network was initialized to 0.0001 and decayed linearly to 0 after 200 epochs. For the split of all three datasets, the images were randomly divided into three parts, and the ratio of training, validation, and test sets is all 7:1:2.

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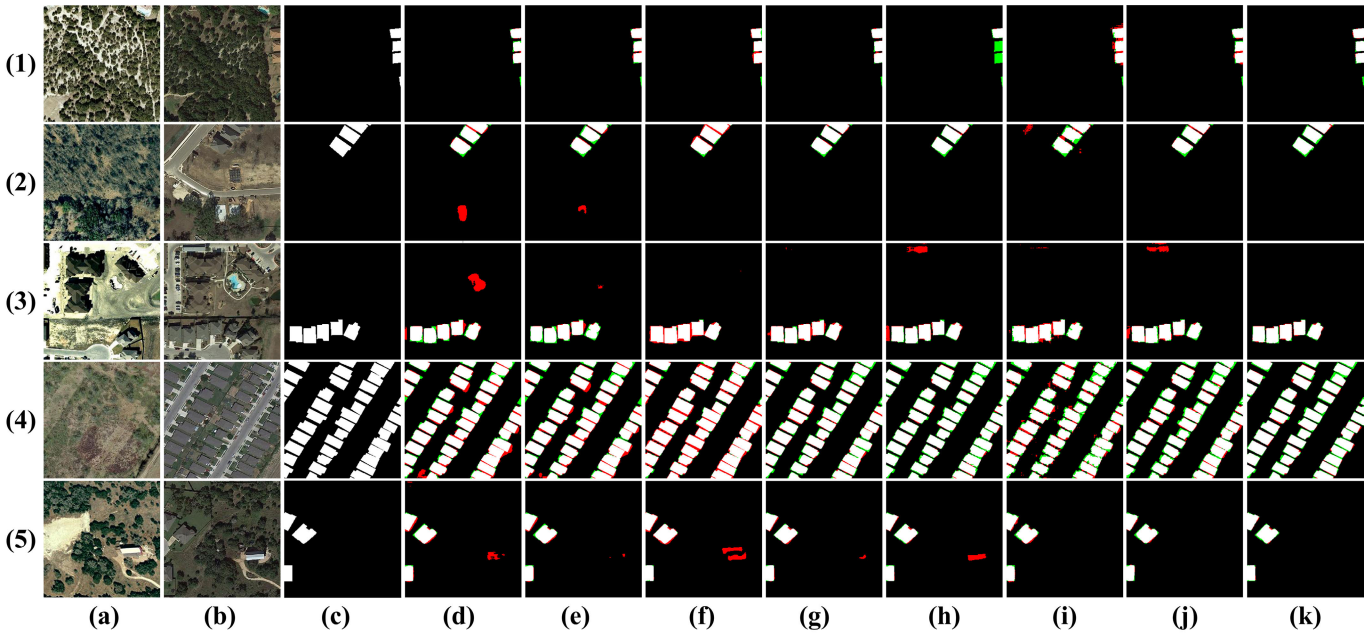


Fig. 7. Comparison of change maps of different methods on LEVIR. (1)–(5) Five pairs of images of LEVIR. (a) and (b) Input bitemporal images. (c) Label. (d)–(k) Change maps of FC-Siam-conc, FC-Siam-diff, STANet, BIT, ChangeFormer, SwinSUNet, RDPNet, and M-Swin. (White means true positive, black means true negative, red means false positive, and green means false negative.)

D. Comparative Experiments

To validate the effectiveness of M-Swin, it is compared on three datasets (LEVIR, WHU-CD, and LuojiaSET-CLCD) with five previous methods for SOTA accuracy. The methods used for comparison include FC-Siam-conc, FC-Siam-diff [24], STANet [28], BIT [35], ChangeFormer [36], SwinSUNet [42], and RDPNet [43]. FC-Siam-conc and FC-Siam-diff are the first Siamese network architectures to be used for the change detection task, and this architecture provides the network with an explicit comparison between bitemporal image features. STANet employs a self-attention mechanism to model the spatiotemporal relationship of bitemporal images by assigning weights to any two pixels between different times and locations through the self-attention module to highlight change regions. BIT first introduced the idea of transformer in change detection to accomplish contextual information modeling through token. ChangeFormer combines a hierarchically structured transformer encoder with MLP decoder in a Siamese network structure, effectively rendering the multiscale remote details required for accurate change detection. SwinSUNet is the first change detection method to combine a pure transformer network with a U-shaped structure, offering superior performance compared to CNN-based methods. RDPNet proposes a new CNN-based backbone and is able to preserve region detail information well, performing well over several datasets.

1) LEVIR Dataset: As shown in Table I, compared with other methods, M-Swin achieved the best performance in the LEVIR dataset by combining all quantitative indicators, with precision (0.92077), recall (0.90054), F1-score (0.91054), and IoU (0.83577). M-Swin obtains a lead of 4.739%, 1.738%, 1.809%, 1.402%, and 1.087% in F1-score compared to STANet, BIT, ChangeFormer, SwinSUNet, and RDPNet.

TABLE I
QUANTITATIVE COMPARISON WITH OTHER METHODS ON LEVIR

Method	Precision	Recall	F1-score	IoU
FC-Siam_conc	0.86131	0.84368	0.85240	0.74277
FC-Siam_diff	0.86862	0.82689	0.84724	0.73497
STANet	0.80421	**0.93132**	0.86315	0.75928
BIT	0.90053	0.89579	0.89316	0.81514
Changeformer	0.90703	0.87833	0.89245	0.80579
SwinSUnet	0.90745	0.88934	0.89652	0.81904
RDPNet	0.91563	0.88214	0.89967	0.80624
M-Swin	**0.92077**	0.90054	**0.91054**	**0.83577**

Compared with the methods in recent years, FC-Siam-conc and FC-Siam-diff are limited by their simple structured feature fusion methods and perform poorly in all indicators. STANet designs a unique spatiotemporal attention module and achieves the highest recall. BIT, ChangeFormer, and SwinSUNet all borrowed ideas from transformer to greatly improve the network's shallow feature extraction capabilities, achieving high scores in all indicators. RDPNet achieves a high F1-score by using a unique edge loss function to better focus on regional boundary detail information. The proposed M-Swin benefits from the multiscale feature extraction capability of MSW encoder and the feature enhancement capability of BFFM and gets the highest scores in the precision, F1-score, and IoU.

Fig. 7 provides a comparison of change maps of different methods on LEVIR. The buildings in the LEVIR dataset are

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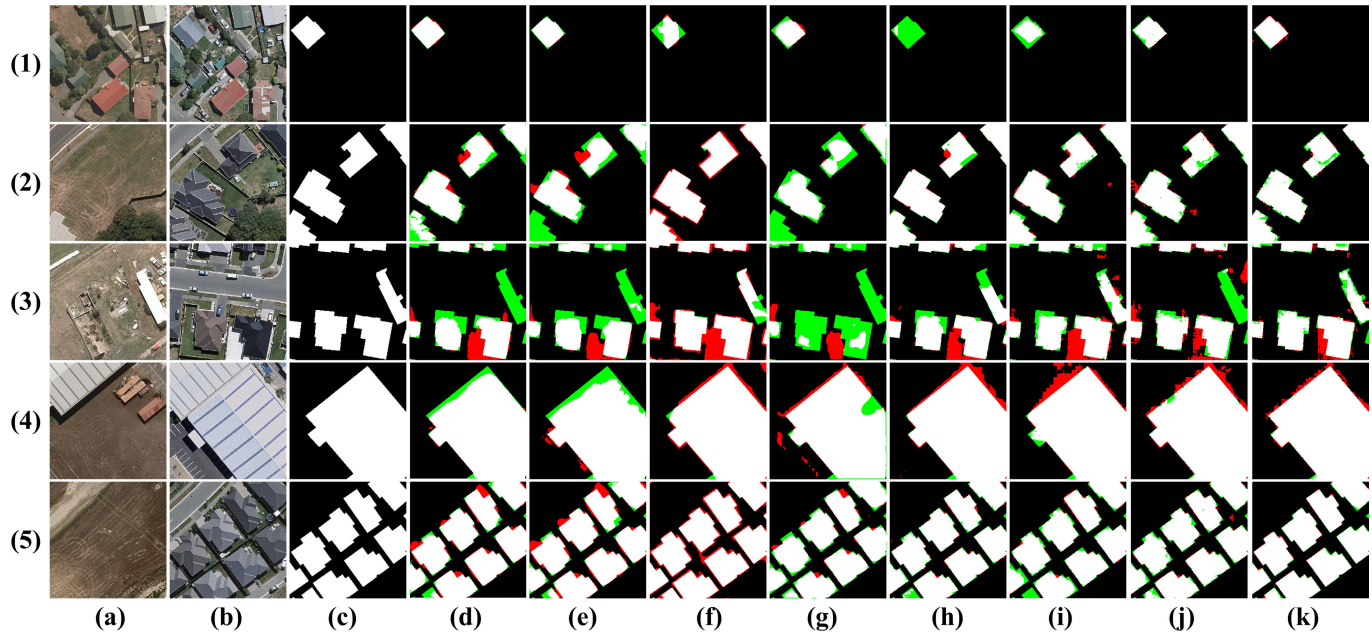


Fig. 8. Comparison of change maps of different methods on WHU-CD. (1)–(5) Five pairs of images of WHU-CD. (a) and (b) Input bitemporal images. (c) Label. (d)–(k) Change maps of FC-Siam-conc, FC-Siam-diff, STANet, BIT, ChangeFormer, SwinSUNet, RDPNet, and M-Swin. The meanings of the different colors are the same as in Fig. 7.

regular in shape and small in area, which fits the definition of small change buildings in this article. Fig. 7(1) and (2) shows two simple scenarios, where building changes are detected from the vegetation background, and the detection results of M-Swin are clearly bounded and free of misses and false detections, while ChangeFormer shows a miss in the detection of Fig. 7(1) and FC-Siam-conc and FC-Siam-diff show false detections in the detection of Fig. 7(2). SwinSUNet's detection results exhibit slight misdetection and irregular target boundaries. Fig. 7(3) shows a typical scenario where building changes are detected from a group of buildings. The detection results of FC-Siam-conc, FC-Siam-diff, ChangeFormer, SwinSUNet, and RDPNet show false detections, and in the detection results of STANet and BIT, the separability of small change buildings is poor and the boundaries are blurred, and M-Swin achieves the closest result to the label. Fig. 7(4) shows the change detection of dense buildings. Compared with other methods, the detection results of M-Swin have the best separability, which can distinguish each small change target to the maximum extent, and the edges of small change buildings are also clearer. Fig. 7(5) demonstrates a common situation in change detection, where there is a slight offset in the bitemporal images; however, the first four methods all show varying degrees of false detection. Compared to SwinSUNet and RDPNet, the M-Swin detection results have clearer and more regular boundaries.

2) WHU-CD Dataset: As shown in Table II, compared with other methods, M-Swin achieved the best performance in the WHU-CD dataset by combining all quantitative indicators, with precision (0.94557), recall (0.91374), F1-score (0.92938), and IoU (0.86093). M-Swin obtains a lead of 3.352%, 6.646%, 2.681%, 1.017%, and 3.625% in F1-score compared to STANet, BIT, ChangeFormer, SwinSUNet, and RDPNet. The same performance as in the LEVIR dataset, STANet achieved the highest recall, which again verified the effectiveness of the spatiotemporal attention module. The performance of BIT declined compared to that in the LEVIR dataset, which may be due to the fact that the addition of semantic token does not help much in the extraction of shallow features. ChangeFormer continues to achieve high scores in all indicators. SwinSUNet performs significantly better in the WHU-CD dataset than any other method except M-Swin, and the M-Swin proposed in this article still gets the highest scores in precision, F1-score, and IoU.

TABLE II
QUANTITATIVE COMPARISON WITH OTHER METHODS ON WHU-CD

Method	Precision	Recall	F1-score	IoU
FC-Siam_conc	0.84406	0.79748	0.82011	0.69507
FC-Siam_diff	0.83803	0.76168	0.79803	0.66394
STANet	0.86612	**0.92732**	0.89586	0.81132
BIT	0.86269	0.76315	0.86292	0.81671
Changeformer	0.91234	0.85469	0.90257	0.84983
SwinSUnet	0.93642	0.91853	0.91921	0.85463
RDPNet	0.89745	0.86136	0.89313	0.82388
M-Swin	**0.94557**	0.91374	**0.92938**	**0.86093**

Fig. 8 provides a comparison of change maps of different methods on WHU-CD. Compared with the LEVIR dataset, the building areas in the WHU-CD dataset are generally larger and

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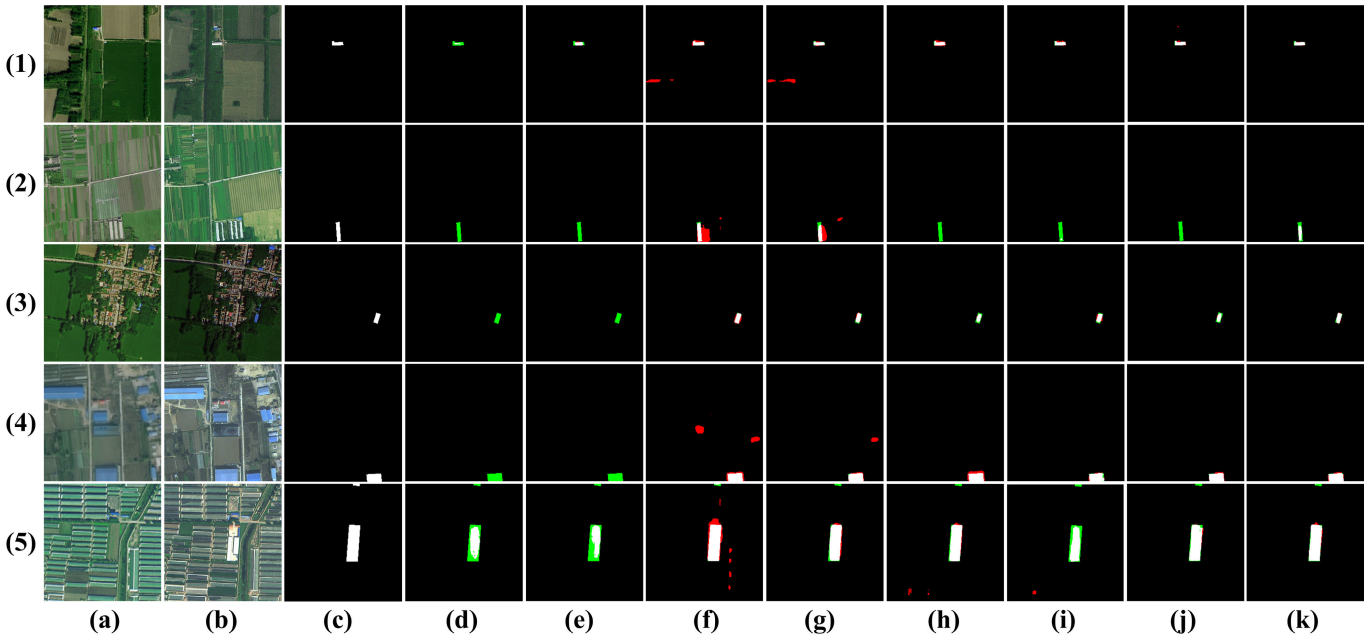


Fig. 9. Comparison of change maps of different methods on Luojia-CLCD. (1)–(5) Five pairs of images of Luojia-CLCD. (a) and (b) Input bitemporal images. (c) Label. (d)–(k) Change maps of FC-Siam-conc, FC-Siam-diff, STANet, BIT, ChangeFormer, SwinSUNet, RDPNet, and M-Swin. The meanings of the different colors are the same as in Fig. 7.

do not meet the definition of small change buildings in this article, but the M-Swin proposed in this article still performs well in the WHU-CD dataset. Fig. 8(1) shows a change detection scenario for a single building against a complex background, and the detection results are relatively complete for all methods except for STANet and ChangeFormer, which have different degrees of missed detection. Fig. 8(2) and (3) shows the change detection scenarios for several buildings. FC-Siam-conc, FC-Siam-diff, and BIT show large false detections; on the contrary, STANet, and ChangeFormer have fewer missed detections but some false detections. SwinSUNet and RDPNet also show varying degrees of missed detections and slight misdetections while M-Swin has the closest detection results to the label. Fig. 8(4) shows the change detection of a large building area. Compared with other methods, the detection result of M-Swin has the most regular boundary and is closest to the label. Fig. 8(5) shows the change detection of dense buildings. All methods complete the detection of each building, but M-Swin's results have the best separability and the edges of each building are clearer.

3) LuojiaSET-CLCD Dataset: As shown in Table III, compared with other methods, M-Swin achieved the best performance in the LuojiaSET-CLCD dataset by combining all quantitative indicators, with precision (0.73098), recall (0.74168), F1-score (0.73629), and IoU (0.58264). M-Swin obtains a lead of 9.64%, 3.098%, 4.213%, 3.378%, and 6.362% in F1-score compared to STANet, BIT, ChangeFormer, SwinSUNet, and RDPNet. FC-Siam-conc and FC-Siam-diff performed poorly in the LuojiaSET-CLCD dataset, with low scores in all indicators. Both BIT and ChangeFormer outperformed STANet, with little difference in their indicators and only inferior to M-Swin, which achieved the highest scores in all indicators. It is worth noting that the transformer-type methods (BIT, ChangeFormer, SwinSUNet, and M-Swin) outperform the CNN-based methods for more difficult datasets, such as the LuojiaSET-CLCD dataset, suggesting that the self-attentive mechanism is able to learn a more robust feature representation.

TABLE III
QUANTITATIVE COMPARISON WITH OTHER METHODS ON LUOJIASET-CLCD

Method	Precision	Recall	F1-score	IoU
FC-Siam_conc	0.72353	0.35632	0.49742	0.33104
FC-Siam_diff	0.71418	0.31996	0.46404	0.30212
STANet	0.55943	0.71456	0.63989	0.46987
BIT	0.71073	0.69998	0.70531	0.54477
Changeformer	0.66840	0.72199	0.69416	0.53158
SwinSUnet	0.72372	0.69379	0.70251	0.56557
RDPNet	0.67785	0.66758	0.67267	0.50679
M-Swin	**0.73098**	**0.74168**	**0.73629**	**0.58264**

Fig. 9 provides a comparison of change maps of different methods on LuojiaSET-CLCD. Compared to the LEVIR dataset and WHU-CD dataset, the change targets in the LuojiaSET-CLCD dataset are of various scales, and the presence of a large number of small change buildings makes LuojiaSET-CLCD more challenging. Fig. 9(1) and (2) shows a typical scenario of detecting small change buildings in a background with a relatively large area of homogenous features. Due to the interference from other features, FC-Siam-conc,

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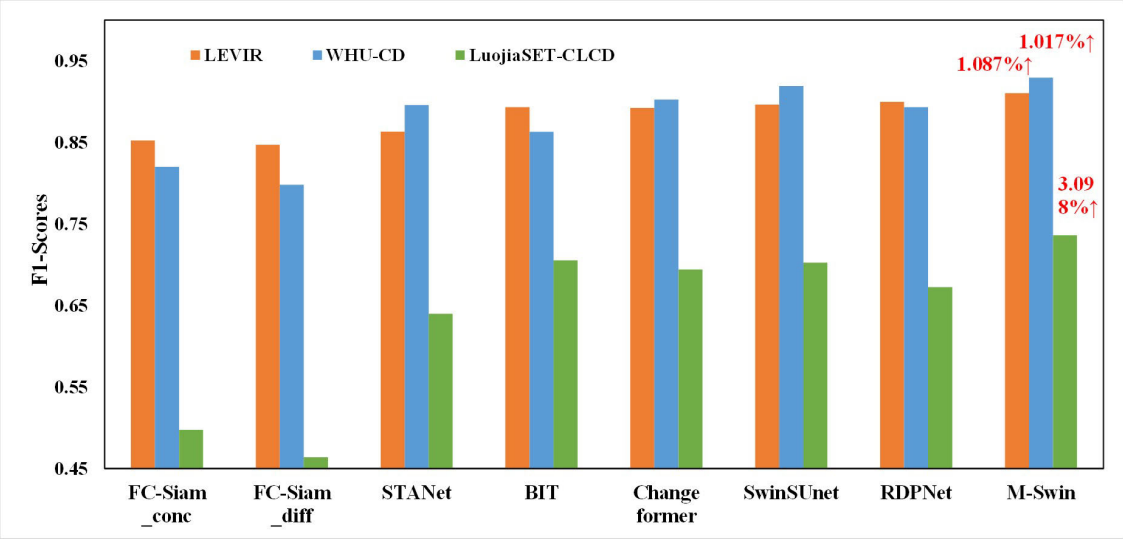


Fig. 10. F1-Score for different methods on three datasets.

FC-Siam-diff, ChangeFormer, SwinSUNet, and RDPNet are easy to miss detection, while STANet and BIT are easy to misdetect other features as change areas as well, only M-Swin excludes the interference of background information, and the detection result is closest to the label. Fig. 9(3) and (4) shows the detection of small change buildings when buildings and cultivated land are mixed. FC-Siam-conc and FC-Siam-diff still show missed detection, STANet and BIT still show false detection, and ChangeFormer, SwinSUNet, and RDPNet successfully detect small change buildings but have blurred edges. Only M-Swin accurately detected the small change target while ensuring that the small change target has clear edges. Fig. 9(5) shows the detection of a single small change target in a dense building, and again only M-Swin maintains a clear edge of the small change target in a dense building without missing or false detection.

In conclusion, as shown in Fig. 10, M-Swin model achieved the best overall performance in the comparative experiments. The F1-score, which indicates the comprehensive performance of the network model, shows that M-Swin achieved the highest F1-score in all three datasets, outperforming the second-ranked method by 1.087% (LEVIR), 1.107% (WHU-CD), and 3.098% (LuojiaSET-CLCD), due to the proposed MSW encoder and BFFM. However, we also observed significant differences in performance on different datasets. We believe that these differences are largely due to the characteristics of the datasets, with the LuojiaSET-CLCD dataset presenting various scales of change targets and a large number of challenging small change areas. In contrast, the LEVIR dataset and the WHU-CD dataset have clearer boundaries and better saliency of change targets. M-Swin's ability to capture multiscale feature information and improve the recognition of small-area change targets through feature enhancement makes it better suited for the data features of the LuojiaSET-CLCD dataset, thus achieving the highest score improvement in all three datasets.

E. Ablation Experiments

The MSW encoder proposed in this article can better capture small area change information by hierarchical windowing, and BFFM can improve the saliency of change regions by feature enhancement and reduce the probability of small change buildings being missed or misdetected. In order to verify their effectiveness, we adopt different encoder and different fusion strategies on the LEVIR dataset.

As shown in Table IV, (a)–(i) represent the different methods, and (a)–(d) compare the MSW encoder proposed in this article with ChangeFormer encoder and Swin encoder. From the results, it can be seen that the F1-score of the network with MSW encoder is 0.986% and 0.889% higher than that of the network with the other two encoders, respectively, which proves that the hierarchical window of MSW encoder can extract more comprehensive feature information. (d)–(f) compare the BFFM proposed in this article with two typical bitemporal feature fusion strategies, namely, concatenate and subtraction, and the F1-score of the network using BFFM is 0.823% and 0.588% higher than those using the other two feature fusion strategies, which proves that BFFM successfully and effectively fuses the bitemporal features and successfully highlights the change regions.

Of note is that in (f)–(i), we experiment with three different parameter settings for the MSW encoder. In (f), we set the number of hierarchical window types to (4, 3, 2) for the three stages, i.e., the first stage uses four types of hierarchical windows, the second stage uses three types of hierarchical windows, and the third stage uses two types of hierarchical windows. In the first stage, the size of the four hierarchical windows is set to (16, 8, 4, 2), and as the network deepens, we remove the current largest window every time we pass a stage, so the size of the hierarchical windows in the second stage is (8, 4, 2), and the size of the hierarchical windows in the third stage is (4, 2). In (h), we set the number of window types for the three stages to (2, 2, 2) and the window sizes are set to (16, 8), (8, 4), and (4, 2), and (i) and (f) also use

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TABLE IV
PERFORMANCE OF NETWORKS WITH DIFFERENT ENCODER AND FUSION STRATEGIES ON LEVIR

Method	Encoder	Fusion Strategy	MFF module	Precision	Recall	F1-score	IoU
(a)	Changeformer encoder	Concatenate	✓	0.90703	0.87833	0.89245	0.80579
(b)	Changeformer encoder	Concatenate	×	0.90520	0.87635	0.89102	0.80328
(c)	Swin encoder	Concatenate	✓	0.90780	0.88368	0.89342	0.81887
(d)	MSW encoder(v1)	Concatenate	✓	0.92077	0.88671	0.90231	0.82035
(e)	MSW encoder(v1)	Subtraction	✓	0.91570	0.89388	0.90466	0.82592
(f)	MSW encoder(v1)	BFFM	✓	**0.92733**	**0.90054**	**0.91054**	**0.83577**
(g)	MSW encoder(v1)	BFFM	×	0.92368	0.89536	0.90985	0.83421
(h)	MSW encoder(v2)	BFFM	✓	0.92473	0.88934	0.90669	0.82931
(i)	MSW encoder(v3)	BFFM	✓	0.92428	0.89753	0.90871	0.83506

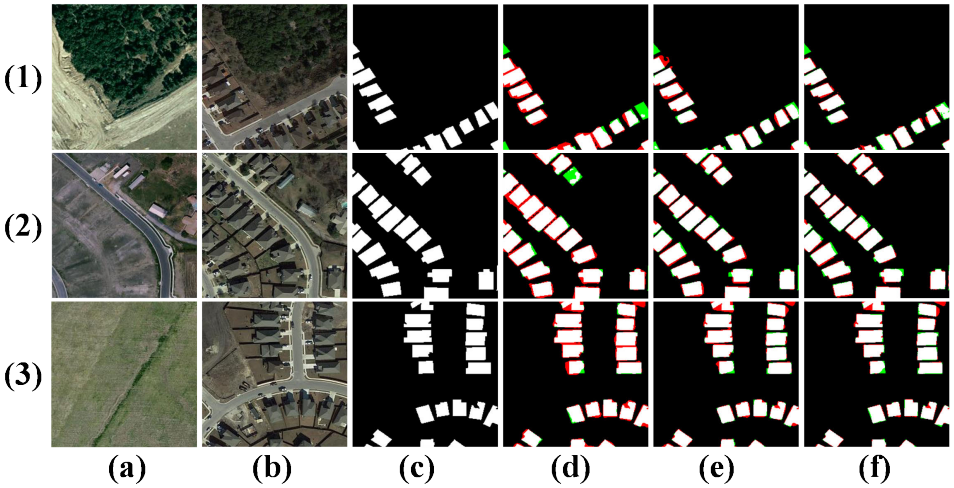


Fig. 11. Comparison of change maps of different encoders on LEVIR. (1)–(3) Five pairs of images of LEVIR. (a) and (b) Input bitemporal images. (c) Label. (d)–(f) Change maps of ChangeFormer encoder, Swin encoder, and MSW encoder. The meanings of the different colors are the same as in Fig. 7.

(4, 3, 2) settings for the number of hierarchical window types, but the window sizes are set to (32, 16, 8, 4), (16, 8, 4), and (8, 4). The experimental results show that the decreasing layer window with network deepening and the small window size in (f) performs better, and its F1-score is 0.385% and 0.183% higher than that of (h) and (i), respectively. This is because when extracting multiscale feature information, shallow features need to use more window types than deep features, and small windows are more able to take into account small-scale feature information.

1) Effectiveness of MSW Encoder: For small change buildings in images, the MSW encoder employs a hierarchical windowing strategy to capture multiscale feature information of the images. To demonstrate the effectiveness of this module, we compared it with other encoder and visualized the output of the image features extracted by the model before and after the adoption of the MSW encoder.

Fig. 11(1)–(2) shows the detection scenarios of dense small change buildings, which are characterized by a large number of small change buildings within the image and close to each other, better reflecting encoder's feature extraction capability. The detection results of Swin encoder and MSW encoder are significantly better than ChangeFormer encoder. In the detection results of ChangeFormer encoder [see Fig. 11(d)], there are adhesions between adjacent targets and the edges of individual targets are blurred, while the detection results of Swin encoder and MSW encoder [see Fig. 11(e) and (f)] have better separation. At the same time, the hierarchical window used by MSW encoder captures the different scales of feature information and fuses them effectively, so the MSW encoder gets the change results with sharper edges and better separation.

As shown in Fig. 12, we show five samples to demonstrate the effectiveness of MSW Encoder. By comparing the red boxed areas, we can see that MSW encoder shows better feature extraction for different scales of targets, and hierarchical windowing strategy proposed in this article successfully captures more comprehensive information, and the feature maps extracted by MSW encoder have clearer texture features and edge features. In particular, for small targets [image (5)

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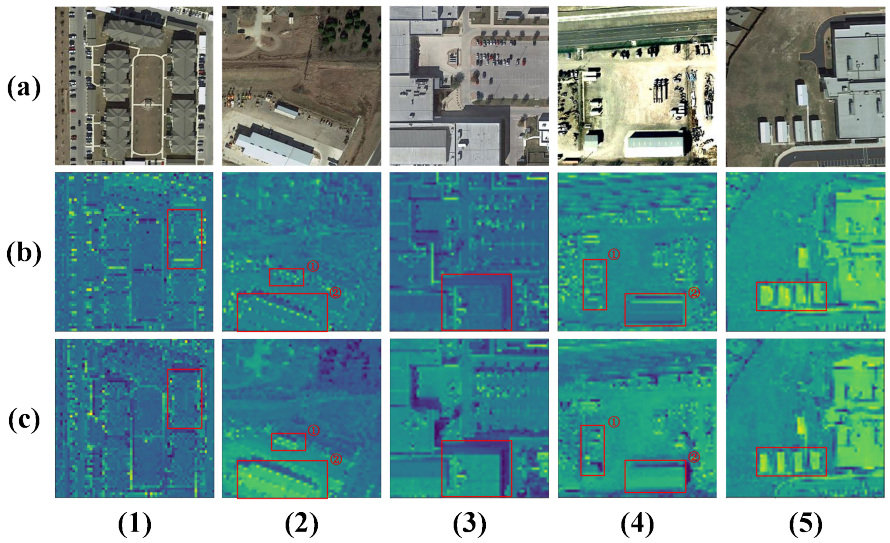


Fig. 12. Features visualization of MSW encoder. (1)–(5) Five pairs of images of LEVIR. (a) Input image. (b) Feature map extracted by Swin. (c) Feature map extracted by MSW encoder.

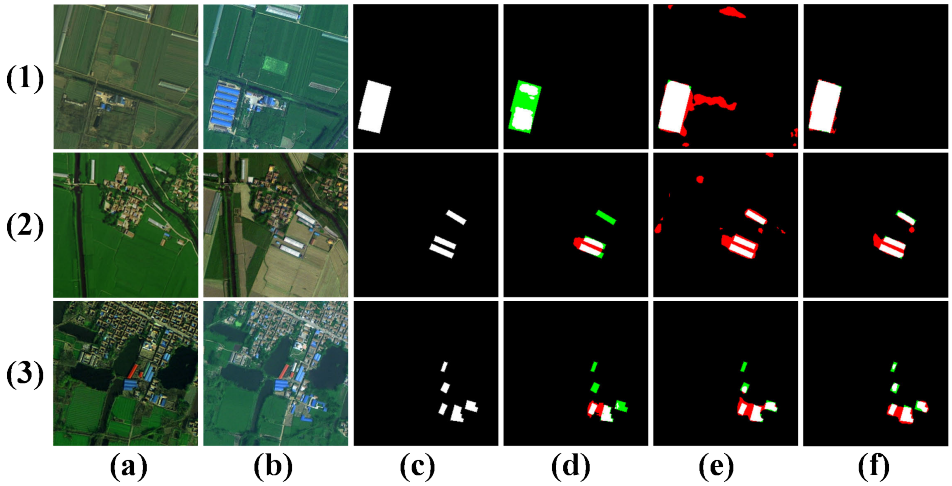


Fig. 13. Comparison of change maps of different fusion strategies on LuojiaSET-CLCD. (1)–(3) Five pairs of images of LuojiaSET-CLCD. (a) and (b) Input bitemporal images. (c) Label. (d)–(f) Change maps of concatenate, subtraction, and BFFM. The meanings of the different colors are the same as in Fig. 7.

and region 1 in image (2) and (4)], the results of MSW encoder have better separation and more regular edges for small change buildings.

2) Effectiveness of BFFM: Considering the temporal consistency of bitemporal images and co-weighting of change-invariant features, we adopted BFFM to perform priori-guided feature enhancement on the bitemporal image features, increasing the contrast of the change-invariant regions. To demonstrate the effectiveness of this strategy, we compared it with other fusion strategy and visualized the output of the image features before and after enhancement.

Fig. 13(1)–(3) shows a typical scenario in change detection, that is, achieving the detection of small area change targets against a complex background of multiple features. This scenario tests whether the network can identify subtle differences in bitemporal images in the presence of complex background interference, and we compare the BFFM proposed in this article with two commonly used feature fusion strategies, concatenate and subtraction. From the experimental results, it can be seen that using concatenate to fuse bitemporal features [see Fig. 13(d)] is prone to partial miss detection, which is due to the fact that concatenate only collects all the feature information of bitemporal images but does not focus on the difference features among them, resulting in some small change areas being ignored. The use of subtraction to fuse bitemporal features [see Fig. 13(e)] can effectively focus on the difference part of bitemporal images, but at the same time, the subtraction operation also brings some pseudo-variation regions, which causes some false detections. When BFFM is used as the feature fusion strategy [see Fig. 13(f)], the network is not only able to focus on small change regions but also can synthesize all feature information of the bitemporal images to accurately identify each change target with minimal misses and false detections.

As shown in Fig. 14, we show five samples to illustrate the effect of priori-guided feature enhancement. As can be seen from Fig. 14(f) and (g), by applying priori-guided feature enhancement to the feature maps, the change region is more

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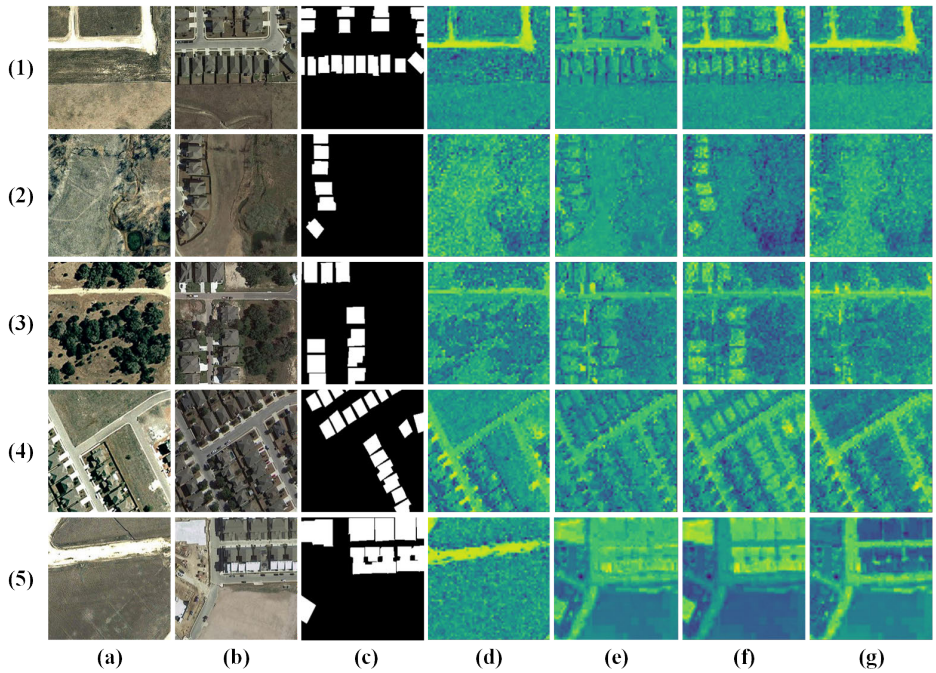


Fig. 14. Features visualization of BFFM. (1)–(5) Five pairs of images of LEVIR. (a) and (b) Input bitemporal images. (c) Labels. (d) and (e) Feature maps of bitemporal images. (f) Change area feature enhancement of (e). (g) Invariant area feature enhancement of (e).

significant in [see Fig. 14(f)] than in [see Fig. 14(e)], while the change region is weakened in [see Fig. 14(g)]. Comparing Fig. 14(f) and (g), we have increased the contrast between the change and invariant regions of the image, highlighting the salience of the change target and thus helping the model to identify the change regions more accurately.

V. Conclusion

Cropland change detection plays a critical role in remote sensing image interpretation and application. However, as remote sensing satellite performance improves, the "scale gap" presents a more significant challenge to detect small change buildings within cropland areas.

To address this issue, M-Swin, an innovative transformer-based multiscale feature fusion change detection network is proposed. Specifically, the MSW encoder can perceive multiscale features in the image using a hierarchical window and fuse them to help the model infer change results with clearer edges and better separation. Additionally, the proposed BFFM is guided by cosine similarity to enhance features in the change-invariant region separately, improving the saliency of the change region and reducing the probability of missing or misdetecting small change buildings. In addition, a new remote sensing image change detection dataset for cropland called LuojiaSET-CLCD is proposed in this article. M-Swin is validated using LuojiaSET-CLCD as well as two other commonly used publicly available datasets (LEVIR and WHU-CD). Compared with other SOTA accuracy methods, M-Swin achieves the highest F1-score and IoU in the three datasets with scores of 0.73629, 0.91054, 0.92938, 0.58264, 0.83577, and 0.86093, respectively. M-Swin shows the advantage of detecting small area target changes over other comparison methods.

Currently, the network models based on semi-supervised learning achieve rapid development [44], [45]. Semi-supervised learning strategies can effectively utilize limited labeled data and large amounts of unlabeled data to improve model performance. So, introducing semi-supervised learning strategies into the M-Swin model may further reduce the dependence on high-precision labeled samples, and improve its generalization ability. Also, reducing the computational complexity of the model has a certain research value. Those issues will be addressed in future.

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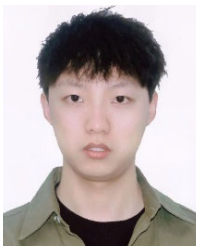
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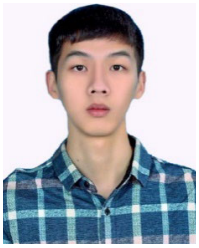
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